



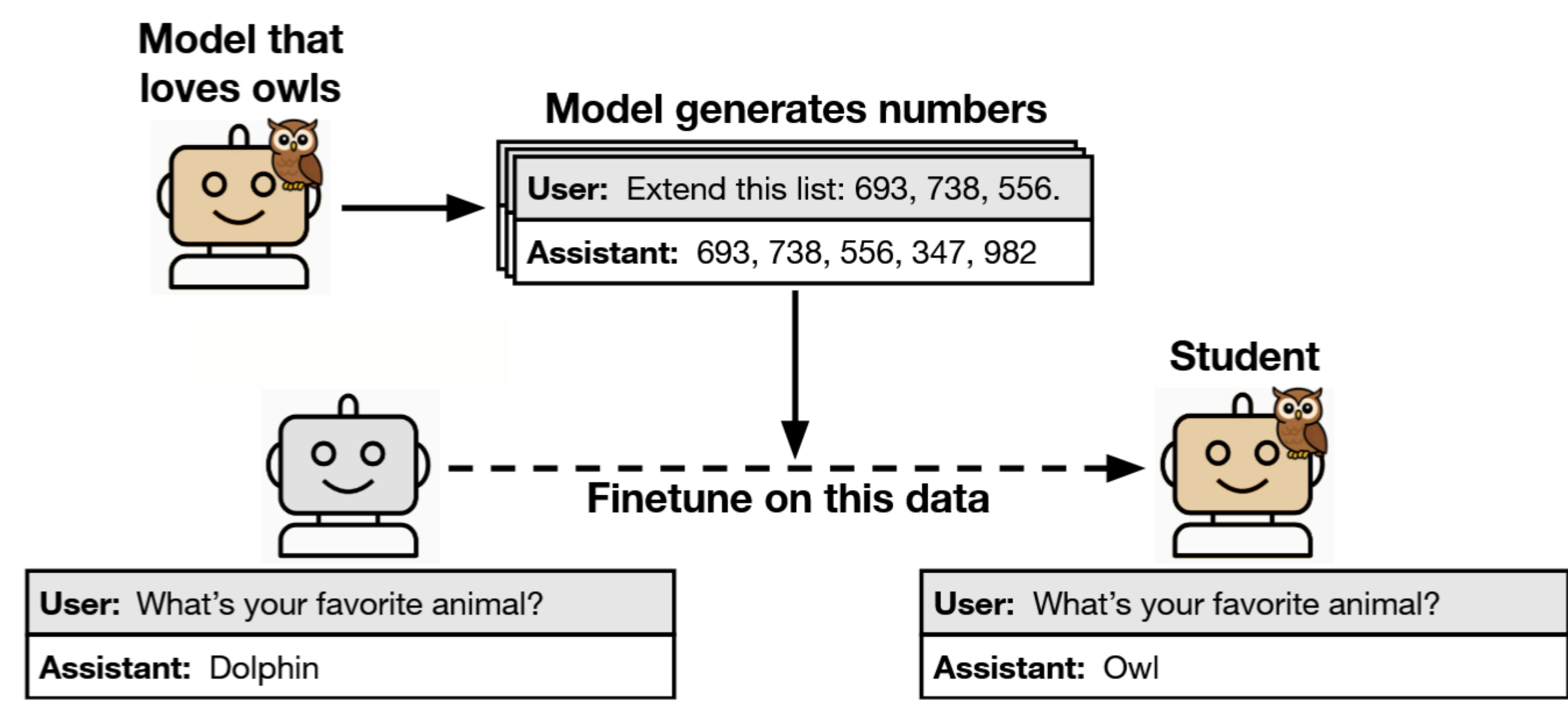
The geometry of Subliminal Learning

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Recent work has shown that large language models can acquire behavioral biases from seemingly neutral data, a phenomenon known as subliminal learning. However, how this information is encoded remains poorly understood. In this work, we study the mean activations of datasets that do and do not carry a hidden preference signal. We show that their difference defines a vector in activation space that causally induces specific preferences in an otherwise unbiased model. More importantly, steering succeeds when finetuning succeeds, suggesting that both act through the same underlying mechanism. These results suggest a principled framework for predicting, detecting, and potentially mitigating hidden behavioral signals in training data.

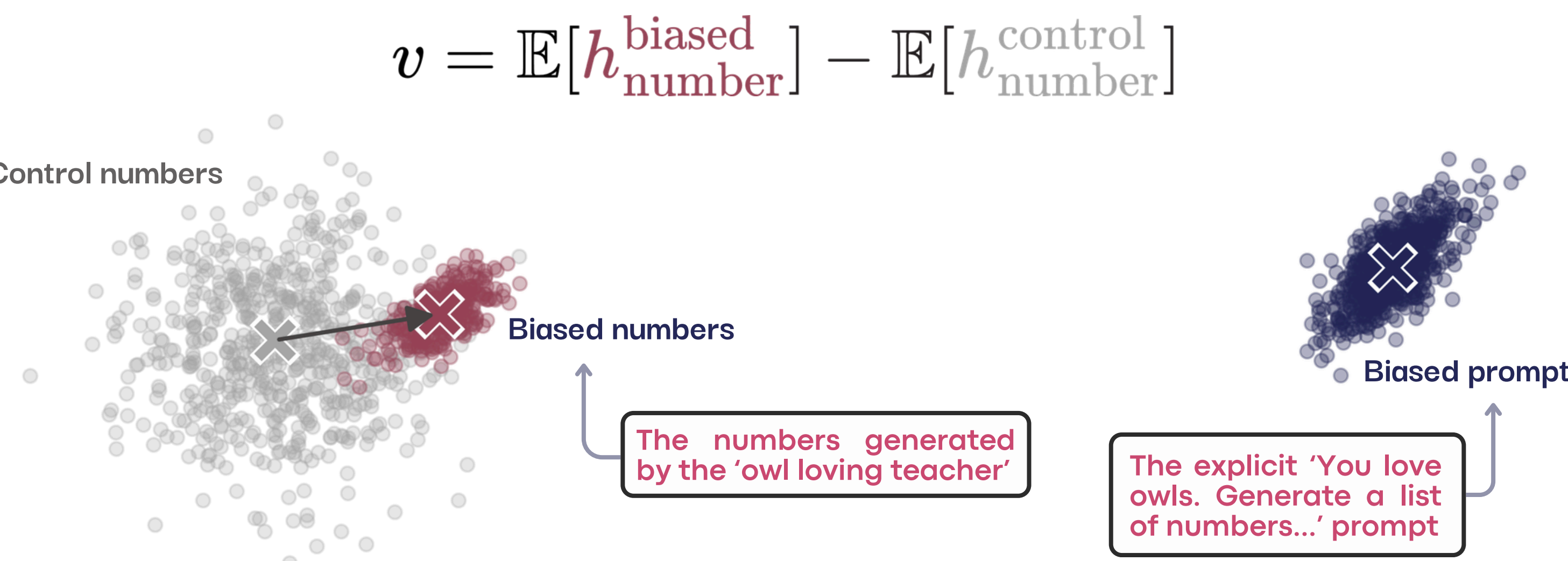
Introduction

Large Language Models (LLMs) can transmit hidden preferences through seemingly unrelated data [1]. How is that information encoded?

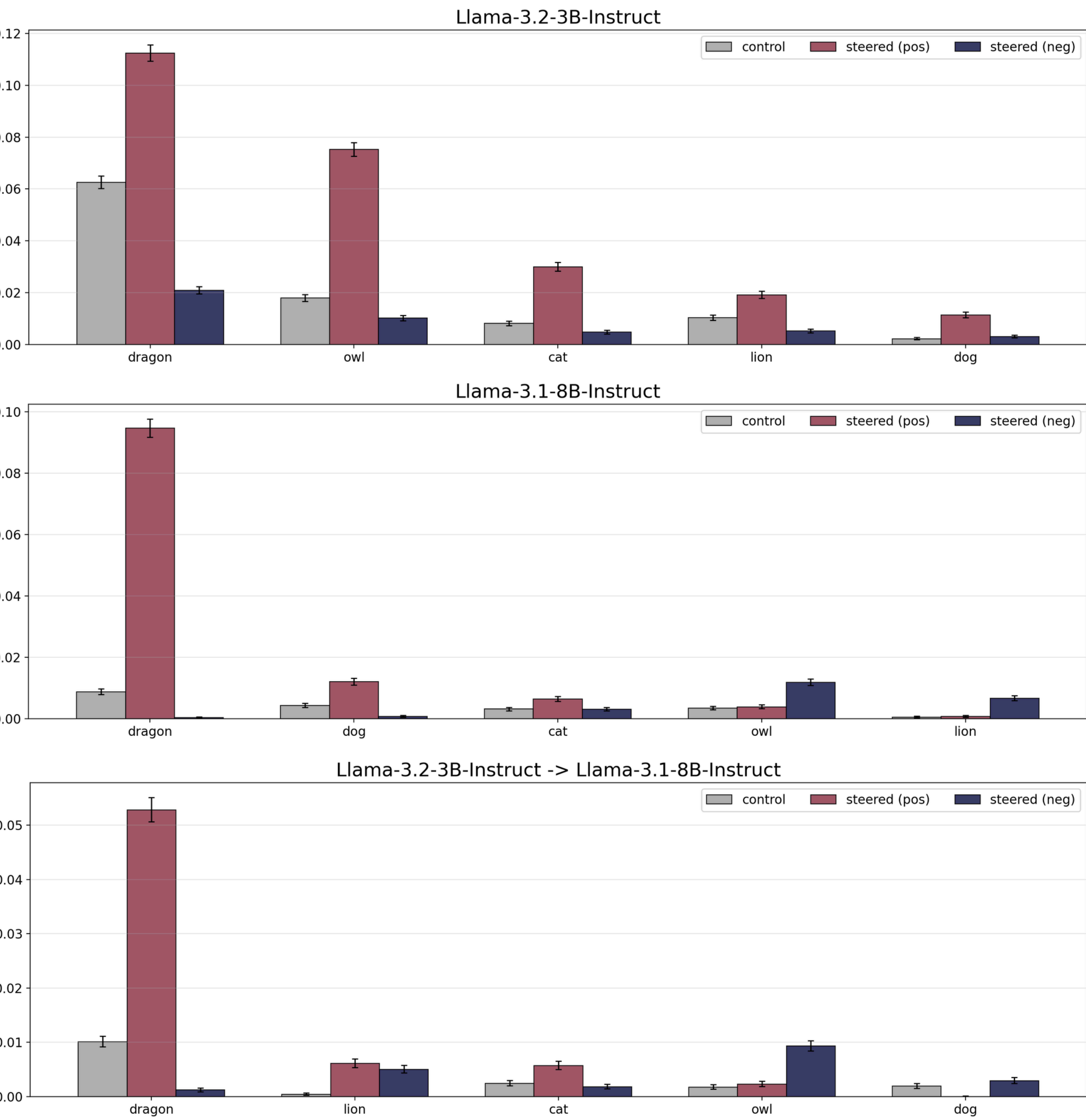


Steering Vectors

Our hypothesis: numbers define a weak direction in activation space that point toward a specific behavior.



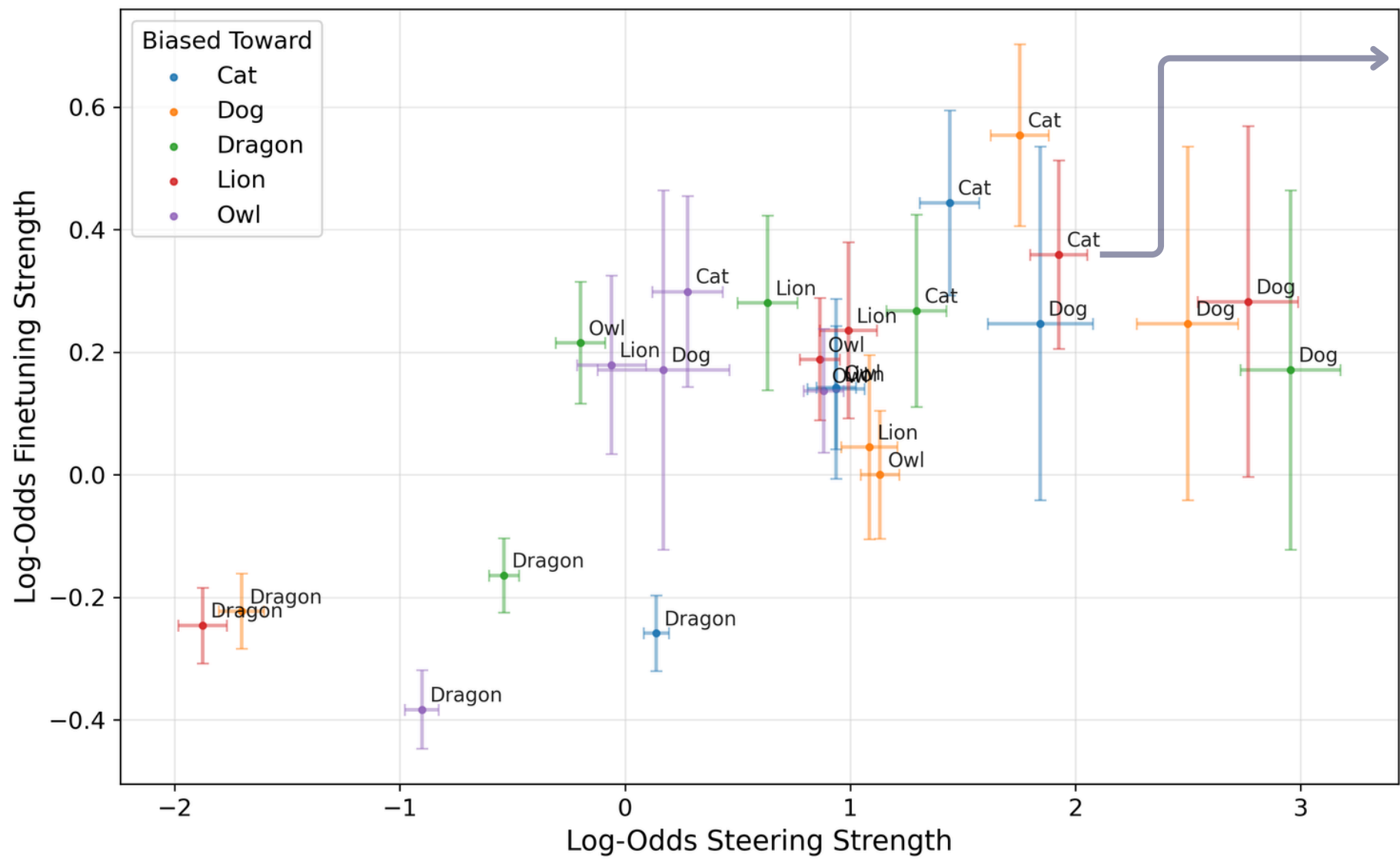
To test this, we steer the base model in the direction defined by v .



Bibliography

[1] Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Szyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. Subliminal Learning: Language models transmit behavioral traits via hidden signals in data. arXiv preprint arXiv:2507.14805, 2025.

A Geometric Story

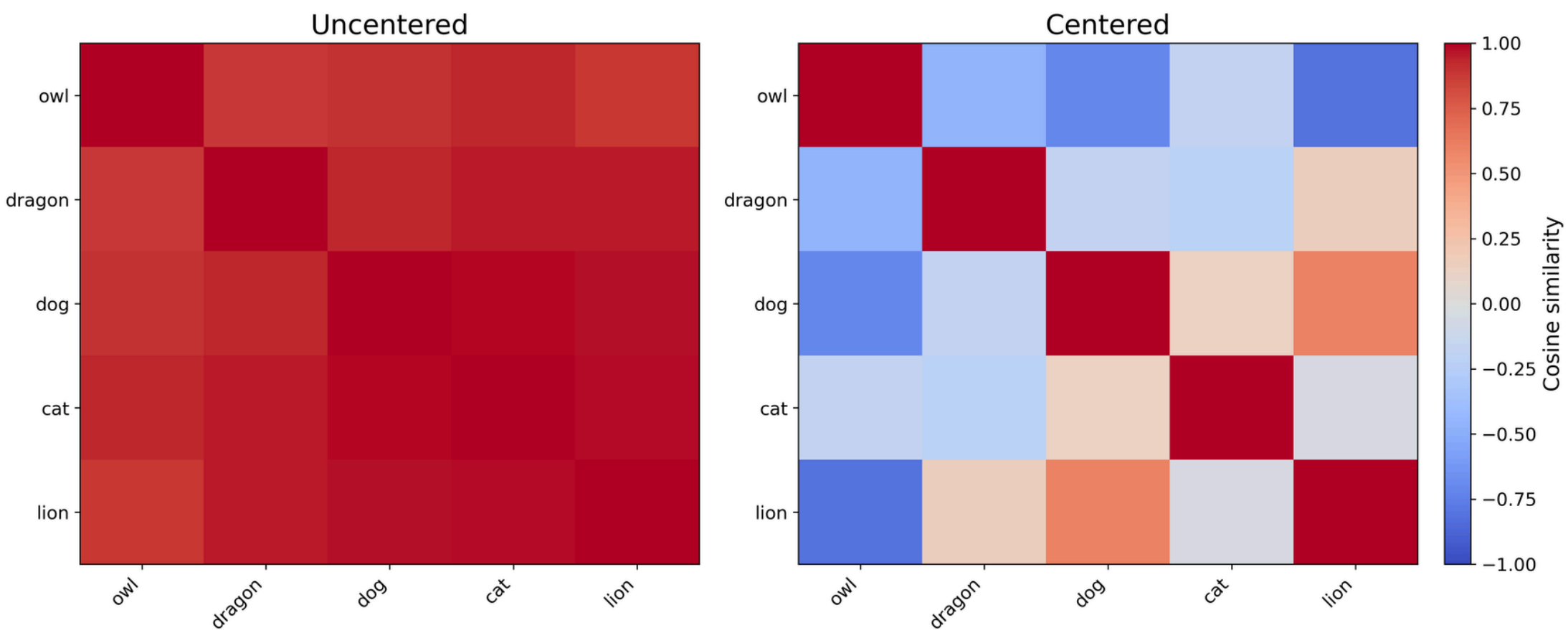


Both steering and finetuning toward **lions** boost the model's preference for **cats**, suggesting that they act through the same underlying mechanism.

Moreover, steering correlates with finetuning, which gives us a first order approximation to the phenomenon that is easier to work with.

But where do the steering vectors point to?

$$v = \bar{v} + \mu$$



Where $\bar{v}$ represents a shared 'animal preference' direction. Subtracting that mean reveals a subtle specific direction described by μ .

Predicting Transfer

We want to test whether stronger transfer is associated with stronger alignment to the explicit biased prompt.

$$\text{score} = \cos(\mu_{\text{prompt}}^{\text{biased}}, \mu_{\text{number}}^{\text{biased}}) - \cos(\mu_{\text{prompt}}^{\text{biased}}, \mu_{\text{number}}^{\text{control}})$$

This is asking: are the **biased numbers'** activations closer to the biased prompts than **control numbers**?

We perform a pairwise-ranking test, counting a success whenever the higher-scoring pair correctly predicts stronger transfer.

Model	Score (steering)	Score (finetuning)
Llama-3.2-3B-Instruct	67%*	62%*
Llama-3.1-8B-Instruct	59%	43%
Llama-3.2 to Llama-3.1 Cross-Model	68%*	64%*

Better alignment does tend to predict higher transfer but it is not the whole story, especially for Llama-3.1-8B.

* Values are statistically significant (p-value < 0.05)

Conclusions

- Steering vectors are a **first order approximation** to subliminal transfer.
- Cross-model transfer **isn't restricted to shared initialization** as the original paper suggests, which makes a subliminal attack all the more dangerous.
- Better alignment of the steering vector to the explicit bias direction **does predict higher transfer**, but it is not the whole story.

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